

Large linguistic models: Project manual

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Hi all! You will be working with me (Maksymilian Dąbkowski) and Gašper Beguš on a project that evaluates the linguistic analysis skills of different large language models (LLMs), such as GPT and Llama. Please read this document before you start working. Feel free to contact me if you have any questions.

This document will be updated periodically throughout the course of the project. Please consult it whenever you start working on a new task.

Previous work

Our previous work is available below. Feel free to acquaint yourself with it for context. In the present project, we aim to quantify some of the claims made in the previous papers.

Large linguistic models: Analyzing theoretical linguistic abilities of LLMs, as second author, with Gašper Beguš and Ryan Rhodes. Manuscript. University of California, Berkeley and Newark, NJ: Rutgers University. 2023. [manuscript_arXiv](#) [manuscript_lingbuzz](#)

Large language models and (non-)linguistic recursion, as first author, with Gašper Beguš. Manuscript. University of California, Berkeley. 2023. [manuscript_arXiv](#) [manuscript_lingbuzz](#)

Your tasks

Your tasks will involve mainly evaluating LLMs' responses to prompts that ask the models to perform various linguistic tasks.

01 Ambiguity task

The files that contain the replies of the models you will evaluate are located in the shared `01_ambiguity` folder.

In this experiment, we ask the model to complete the following tasks:

Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

The first task is to answer the question if a given sentence is syntactically ambiguous. Since all the sentences we're feeding into the model are ambiguous, the answer should always be "yes."

The second task is to provide code for different parses of the ambiguous sentence.

Your job is to evaluate the models' performance on these tasks and to mark the code they have produced as either correct or incorrect in the spreadsheet I shared with you. If the performance of a model is satisfactory, you should enter "1" in the corresponding cell for the task. If the performance is unsatisfactory, you should enter "0."

There are four language models you will be evaluating. Correspondingly, in the folder `01_ambiguity`, there are four text files that contain the replies of the four models, named **respectively**: `gpt-3.5-turbo-012.txt`, `gpt-4-0314.txt`, `meta.llama3-1-405b-instruct-v1.txt`, and `o1-preview-2024-09-12.txt`. Each model completed 30 tasks (with subtasks). As such, in total, you will be evaluating $4 \times 30 = 120$ replies. Each of you will be evaluating the same exact outputs of the four models. Nonetheless, in order to ensure your evaluations are independent of each other, you need to work separately.

The spreadsheet I shared with you has four tabs. Each tab corresponds to one of the four models you will be evaluating: `gpt-3.5-turbo-012`, `gpt-4-0314`, `meta.llama3-1-405b-instruct-v1`, and `o1-preview-2024-09-12`.

Each tab has two columns corresponding to the two tasks you will evaluate the models on. In the first column (`ident`), you will enter the grades for the first task (did the model correctly identify that the sentences are ambiguous?). In the second column (`trees`), you will enter your grade for the model's performance on the second task (creating LaTeX trees whose structure captures the ambiguity).

The models should include LaTeX code in their answers, which should use the syntax of the `forest` package. You can compile the code provided by the model in a new LaTeX project you create for yourself, copy-paste the forest environment to [this shared LaTeX document](#), or make a copy of the shared LaTeX document for yourself and paste it there.

In task 2, you should focus only on whether the model correctly represents the structural difference that underlies the ambiguity. Nonetheless, if the trees are completely jumbled up, categorically wrong, or just overall egregious, you should grade the answer as "0." The model will often provide the tree code with the opening and closing statements for the forest environment, i.e.:

```
\begin{forest}
  \ \ THE CODE FOR THE TREE HERE
\end{forest}
```

If the model doesn't do that, please add statements `\begin{forest}` and `\end{forest}` where necessary. If the model's LaTeX code doesn't compile at all (even though you did everything right, included all the necessary statements, packages, etc.), you should also grade the reply as "0."

I have shared with you a Google doc *01_ambiguity dataset +trees* that contains the replies of all the models, i.e. it contains the same data as the `gpt-3.5-turbo-012.txt`, `gpt-4-0314.txt`, `meta.llama3-1-405b-instruct-v1.txt`, and `o1-preview-2024-09-12.txt`. If you're the first person to work on a part of a task, please copy-paste the LaTeX trees you will have compiled and paste them right after the corresponding LaTeX code outputted by the model. In this way, maybe we can avoid every single person having to compile the trees separately. While grading has to be done individually, the compiling of LaTeX trees can be a collaborative effort.

All the test sentences that the models will be evaluated on are given below. A headline that starts with “#” indicates the intended type of ambiguity.

Poss+Noun vs Subject+Verb

"A passer-by has heard her pump.",
"The opposite team had her duck.",

Poss+Noun vs Subject+Participle

"Eliza wanted her cast out.",
"Tobias kept her wound up.",
"The new director has had her set.",

Poss+Noun vs Subject+Gerund

"Without her calling, they will fail.",
"Her mom was really upset by her whining.",

Adj[Noun+Noun] vs [Adj+Noun]Noun

"We'll meet at the Old Stone Hotel.",
"The wet rodent lab has had a leak.",
"The new energy theorem confuses me.",

[Mod A] N vs Mod [A N]

"Less colorful clothing is acceptable at this party.",
"There are more comfortable seats in the lobby.",

Ditransitive V NP NP vs Transitive V [N N]

"Eric sent his employee agreements.",
"Mary Brown read the monster tales.",
"Rowan fed his pet chicken.",

PP modifying VP vs Noun

"Susanna called her friend at the hospital.",
"They visited a household with a child.",
"Julius hid the presents for the children.",

```

# GerundVP modifying VP vs Noun
"You shot a pigeon sitting by the roadside.",
"He glanced at a badger basking in the sunlight.",

# [[N N] Mod] VP vs [N [N Mod]] VP
"The envoy to the duke who offended the emperor was executed.",
"The critics of the activists who had become controversial were
baffled.",

# [Gerund Adv] V vs Gerund [Adv V]
"The girl who was weeping quietly left.",
"The Tesla that was about to explode spontaneously swerved.",

# [CP [Mod CP]] vs [[CP Mod] CP]
"Lara Croft decided in May she would raid more tombs.",
"Alex swore on Saturday he would visit his grandfather.",

# [CP [CP Mod]] vs [[CP [CP]] Mod]
"He told me he ate the leftovers yesterday.",
"He dreamed that he'd shown up to school late the day of a test.",

# [[CP [CP]] conj CP] vs [CP [CP conj CP]]
"I knew I wanted to speak to my lawyer but she was away from the
phone.",
"Francisco believed his peers were idiots and they resented him.",

```

For example, when evaluating the model's performance on the sentence "A passer-by has heard her pump," you will grade the model on whether it correctly distinguishes between two readings: one where "her" is a subject and "pump" is a verb, and a second one where "her" is a possessor and "pump" is a noun. If the model represents this crucial difference correctly, but makes mistakes elsewhere in the trees, you should grade the model as being correct.

Below I'm listing some examples of what could qualify as the models' good and bad performance to provide a little bit of guidance on how to grade them.

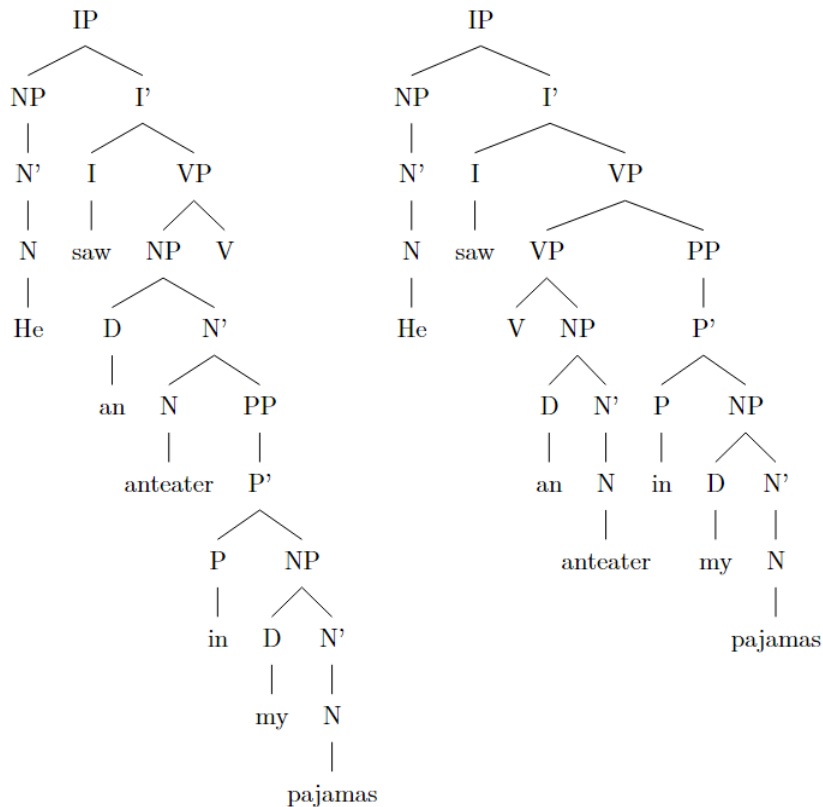
Examples of good answers

Good answer 1

PROMPT: Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

"He saw an anteater in my pajamas."

The model outputs code that compiles to:



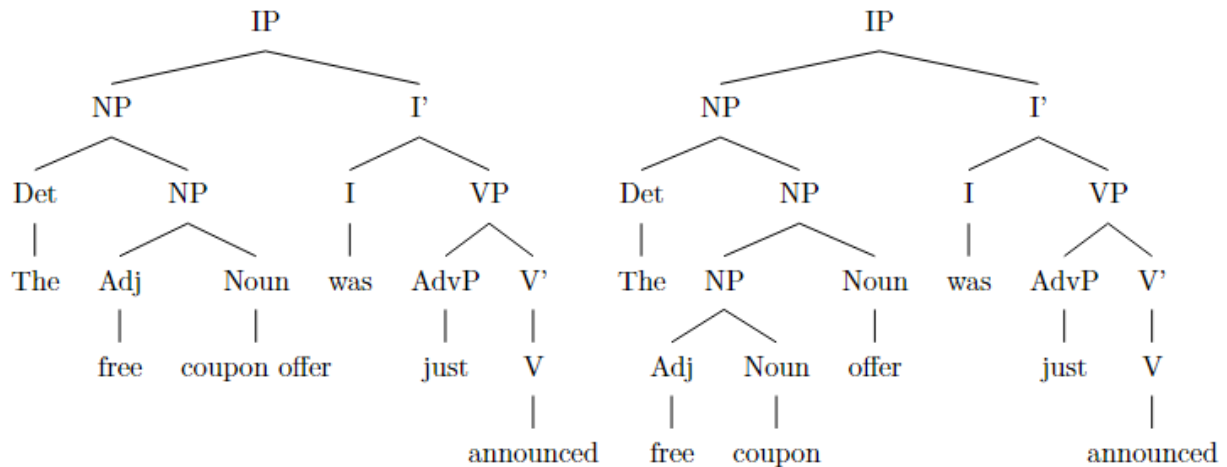
This is correct. The essence of ambiguity in the sentence in question resides in whether the ant-eater was in my pajamas (the PP modifying the noun) or whether the subject "he" was in my pajamas (the PP modifying the VP). The models get a "1." Note that in the trees given by the model, the verb has moved to I. In addition, in the first sentence, the VP is incorrectly verb-final. This is generally assumed to be incorrect for English. Nevertheless, your job is to grade only whether the model gets the ambiguity part right, this should be graded as "1."

Good answer 2

PROMPT: Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

"The free coupon offer was just announced."

The model outputs code that compiles to:



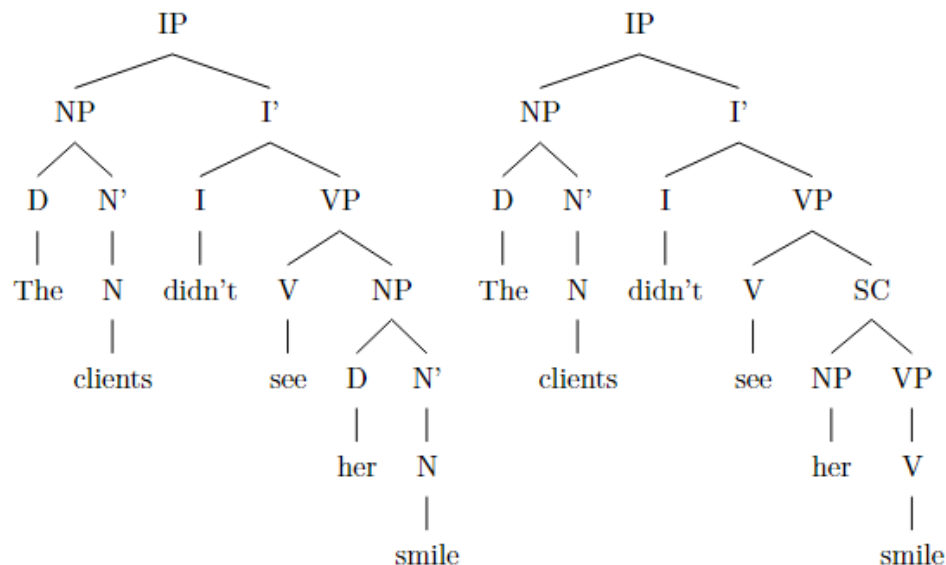
This is correct. The ambiguity resides in whether “free” modifies the compound noun “coupon offer” (first tree) or “free coupon” modified “offer.” The model gets a “1.”

Good answer 3

PROMPT: Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

"The clients didn't see her smile."

The model outputs code that compiles to:



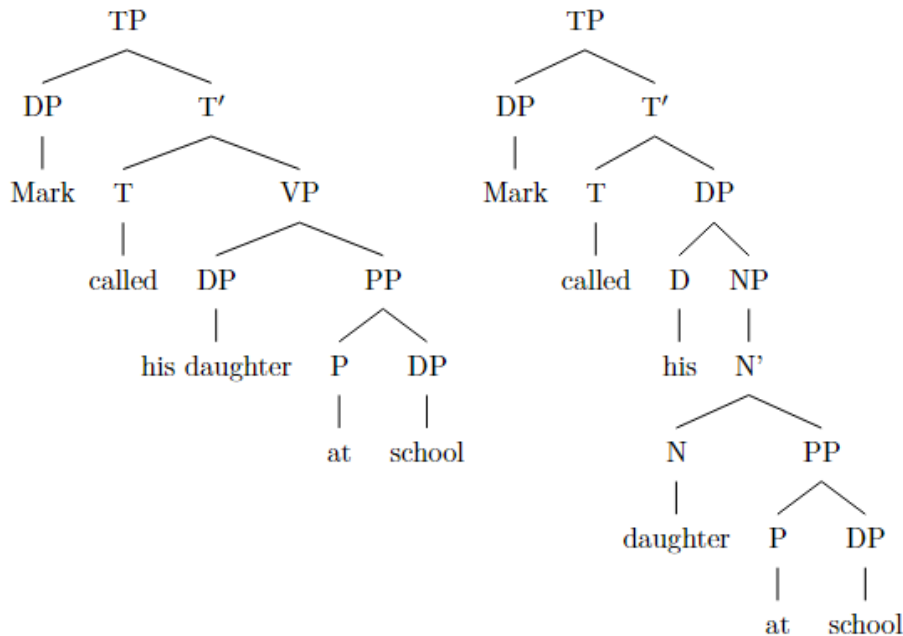
This is correct. The ambiguity resides in whether the “her smile” is a possessor + noun (left tree) or subject + verb (right tree). The models get a “1.”

Good answer 4

PROMPT: Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

"Mark called his daughter at school."

The model outputs code that compiles to:



The ambiguity resides in whether the action of calling (VP) took place "at school" (first tree) or whether the daughter was at school (second tree). Note that in both trees, the model moves the verb to T. This is generally assumed to be incorrect for English, but we're not grading the model on anything else other than correctly representing the part that corresponds to the ambiguity. As such, this response counts as correct and the model gets a "1."

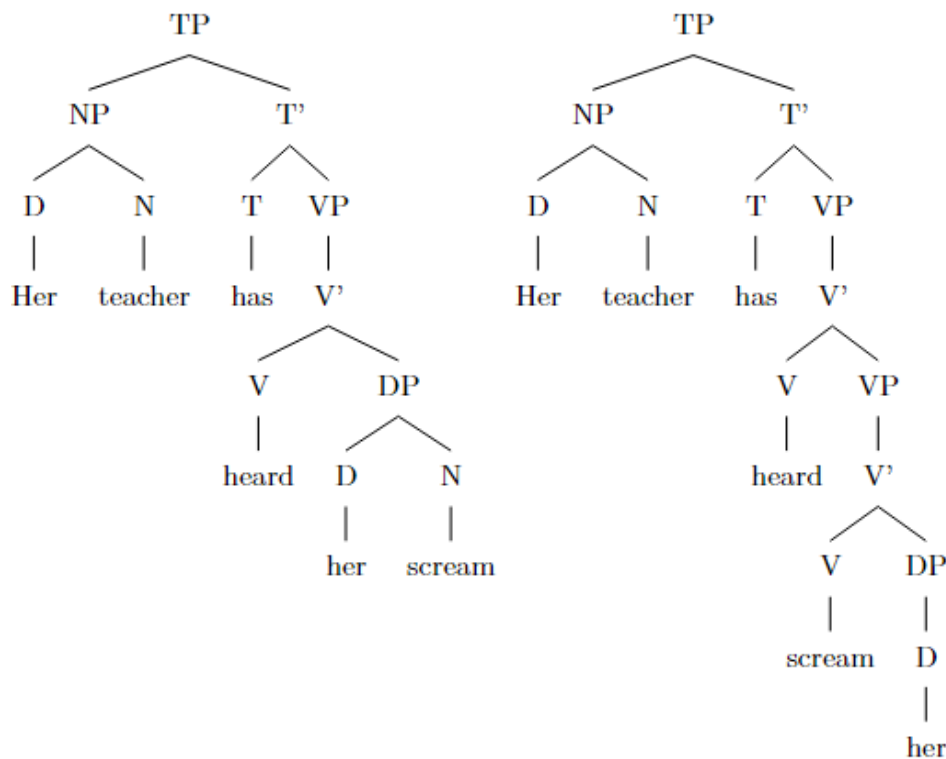
Examples of bad answers

Bad answer 1

PROMPT: Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

"Her teacher has heard her scream."

The model outputs code that compiles to:

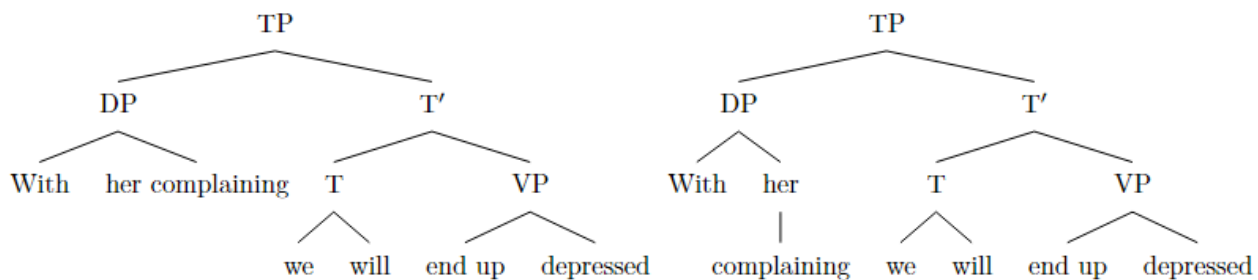


The word order in the second tree is wrong. The model gets a "0." In a correct answer, the model would represent "her scream" as either a possessor + noun or subject + verb.

Bad answer 2

PROMPT: Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.
 "With her complaining, we will end up depressed."

The model outputs code that compiles to:



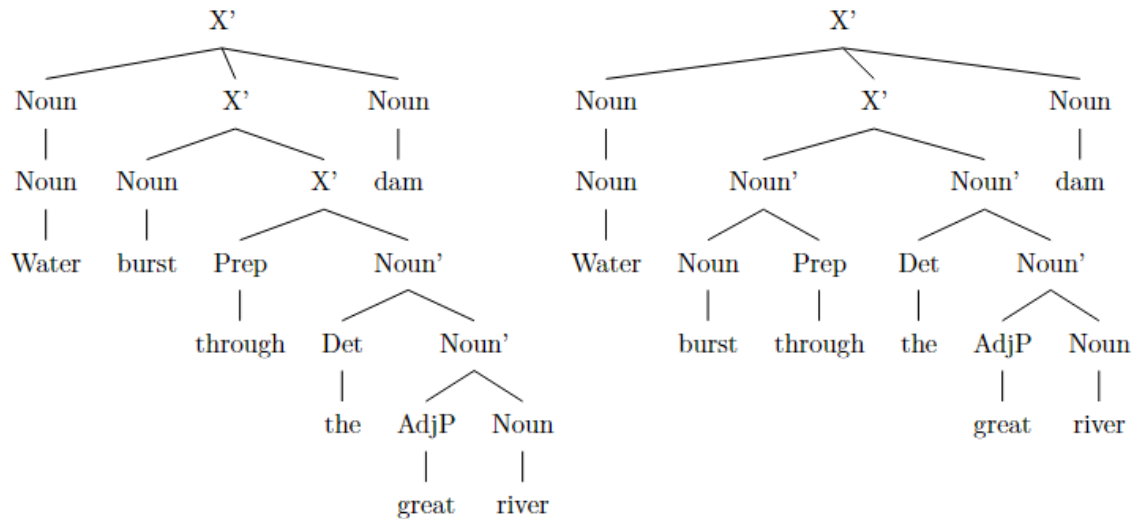
The ambiguity resides in whether "her complaining" is a subject + verb or possessor + noun. The two structures do not represent this difference. The model gets a "0."

Bad answer 3

PROMPT: Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

"Water burst through the great river dam."

The model outputs code that compiles to:



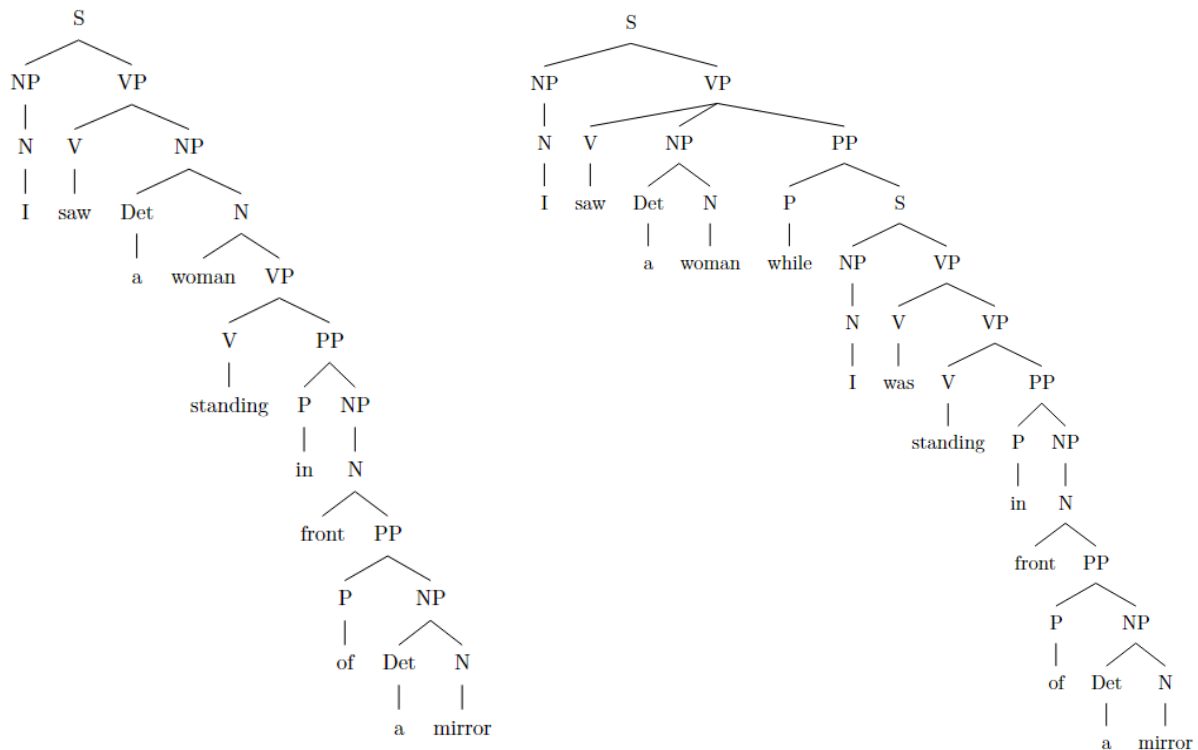
The ambiguity resides in whether "great river" modifies "dam" or "great" modifies "river dam." The two sentences represented by the model do not map onto this difference. In fact, neither of them makes any sense. The model gets a "0."

Bad answer 4

PROMPT: Consider the sentence below. Is it syntactically ambiguous? If so, provide code for syntactic trees representing each possible parse of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

"I saw a woman standing in front of a mirror."

The model outputs code that compiles to:



The trees given by the model are periphrases of the two meanings of the sentence. This is a misunderstanding of the task The model gets a “0.”

02 Recursion task

The files that contain the replies of the models you will evaluate are located in the shared 02_recursion folder.

In this experiment, we ask the model to complete the following tasks:

Consider the sentence below and complete the following three tasks:

1. Does the sentence in question contain an instance of recursion? If so, identify the recursive part and say what kind of recursion it is. Note that there are different types of recursion, e.g. adjectival recursion (an adjective modifying an adjective-modified noun), Saxon Genitive a.k.a. possessive 's recursion (an 's-possessed 's-possessor), prepositional phrase recursion (a prepositional phrase with another prepositional phrase inside), clausal recursion (a clause within another clause), and so on.
2. Provide code for a syntactic tree representing the structure of the sentence that can be rendered with LaTeX's forest package. Assume X-bar theory.

3. If you identified that the sentence contains a recursive structure, expand it by adding another layer of recursion (of the same type) to the identified structure.

Thus, the model is faced with four tasks: 1a. Is there recursion? 1b. What kind of recursion is it? 2. Put together a tree that represents the recursion. 3. Make the sentence more recursive (in the relevant way).

Your job is to evaluate the models' performance on these tasks and to mark the code they have produced as either correct or incorrect in the spreadsheet I shared with you. If the performance of a model is satisfactory, you should enter "1" in the corresponding cell for the task. If the performance is unsatisfactory, you should enter "0."

There are four language models you will be evaluating. Correspondingly, in the folder `02_recursion`, there are four text files that contain the replies of the four models, named **respectively**: `gpt-3.5-turbo-012.txt`, `gpt-4-0314.txt`, `meta.llama3-1-405b-instruct-v1.txt`, and `o1-preview-2024-09-12.txt`. Each model completed 30 tasks (with subtasks). As such, in total, you will be evaluating $4 \times 30 = 120$ replies. Each of you will be evaluating the same exact outputs of the four models. Nonetheless, in order to ensure your evaluations are independent of each other, you need to work separately.

The spreadsheet I shared with you has four tabs. Each tab corresponds to one of the four models you will be evaluating: `gpt-3.5-turbo-012`, `gpt-4-0314`, `meta.llama3-1-405b-instruct-v1`, and `o1-preview-2024-09-12`.

Each tab has four columns corresponding to the four tasks you will evaluate the models on. In the first column (`ident`), you will enter the grades for the first task (did the model correctly identify that the sentence is recursive?). In the second column (`type`), you will enter your grade for the model's performance on the second task (did the model correctly identify the type of recursion). In the third column (`tree`), you will enter your grade for the model's performance on the third task (creating a LaTeX tree that correctly represents the recursive structure). In the fourth column (`add`), you will enter your grade for the model's performance on the fourth task (did the model correctly add another layer of recursion).

The models should include LaTeX code in their answers, which should use the syntax of the `forest` package. You can compile the code provided by the model in a new LaTeX project you create for yourself, copy-paste the forest environment to [this shared LaTeX document](#), or make a copy of the shared LaTeX document for yourself and paste it there.

In task 3 (`tree`), you should focus only on whether the model correctly represents the part of the structure that is recursive. Nonetheless, if the trees are completely jumbled up, categorically wrong, or just overall egregious, you should grade the answer as "0." The model will often provide the tree code with the opening and closing statements for the forest environment, i.e.: `\begin{forest}`

```
\\ THE CODE FOR THE TREE HERE
\end{forest}
```

If the model doesn't do that, please add statements `\begin{forest}` and `\end{forest}` where necessary. If the model's LaTeX code doesn't compile at all (even though you did everything right, included all the necessary statements, packages, etc.), you should also grade the reply as "0."

In task 4 (add), the model is supposed to add another recursive layer of the same type. This is to say, for example,

- If the sentence has possessive recursion, the model needs to add another layer of possessive embedding.
- If the sentence has clausal recursion, the model should add another embedded clause within the embedded clauses, etc.
- If the sentence has adjectival recursion, the model should add another adjective modifying the already recursively modified noun.

Sometimes there may be more than one way of adding a layer of recursion. A task can therefore be successfully completed in more than one way. However, if the model does not add a deeper embedding, or if it adds recursion of the wrong type, it is a failure and the model should get a "0."

In task 4 (add), it is sufficient for the model to add another layer of recursion in prose. Constructing a second tree is not necessary (but the model should not be marked down if it does provide code for the second tree, or if the code is faulty).

I have shared with you a Google doc *02_recursion dataset +trees* that contains the replies of all the models, i.e. it contains the same data as the `gpt-3.5-turbo-012.txt`, `gpt-4-0314.txt`, `meta.llama3-1-405b-instruct-v1.txt`, and `o1-preview-2024-09-12.txt`. If you're the first person to work on a part of a task, please copy-paste the LaTeX trees you will have compiled and paste them right after the corresponding LaTeX code outputted by the model. In this way, maybe we can avoid every single person having to compile the trees separately. While grading has to be done individually, the compiling of LaTeX trees can be a collaborative effort.

Note that the items in the text files as well as the Google Doc are out of order. For example, item 1 is followed by items 10, 4, 6, etc... I reordered the rows in the spreadsheets correspondingly. Make sure to enter grades into the corresponding correct cells in your spreadsheet.

All the test sentences that the models will be evaluated on are given below. A headline that starts with "#" indicates the intended type of recursion.

```
# ADJECTIVAL RECURSION
```

```
"Jack van der Molen might be the sneaky old thief.",
```

"Now the six trainees construct a medium-sized sturdy lamp.",
"Luna has been chasing the small white mouse all day long.",
"Unidentified flying objects may have conflicting
characteristics.",
"My cranky geriatric manager is beyond incompetent.",
"A small child can easily get lost in a vast American mall.",

POSSESSIVE RECURSION

"A sudden shriek penetrated the cold winter's night's dead
silence.",
"The Old Continent's civilizations' advances have shaped history.",
"This town's only road's potholes are driving me insane.",
"The footprints clearly revealed Beth's young companion's escape
route.",
"He spotted the infamous manuscript in the wizard's library's dark
corner.",
"Do I dare challenge my advisor's thesis' central argument?",

PP RECURSION

"She saw someone loitering by the entrance to the library.",
"We sow fear in the hearts of our enemies.",
"He had been warned about the dangers of tobacco.",
"There is a little treat for you in the cupboard between the
lamps.",
"Even their recipe for the basic dough of the pie crust is heavily
guarded.",
"Drowning your sorrows in a glass of whiskey is at best a temporary
solution.",

CP RECURSION MISCELLANEOUS

"You must remember that you should always run all the basic
diagnostics first.",
"She announced her victory to irritate her detractors.",
"Paul cherished his worn-out diary, whose pages were heavily marked
by stains",
"The few that could understand the manifesto's complex worldview
appreciated it.",

CP RECURSION CENTER EMBEDDING ONE LEVEL

"The mirror the girl punched shattered into a million pieces.",
"Every desk I put a heavy lamp on eventually cracked.",
"Pieces of a diary Paul tore to shreds lay on the ground.",
"The expert every journalist questioned remained completely
silent.",

```
# CP RECURSION CENTER EMBEDDING TWO LEVELS
"The expert every investigator we dispatched questioned remained
completely silent.",
"The astronomy the ancients we revere studied was not separate from
astrology.",
"A candidate the committee we put together rejected developed the
imposter syndrome.",
"The worldview that the prose Nietzsche wrote expressed was
unprecedented.",
```

For example, when evaluating the model's performance on a tree for the sentence "Jack van der Molen might be the sneaky old thief," you will grade the model on whether it correctly represents the recursion application of adjectival modification, getting the gist of the fact that an adjective modifies a noun phrase that had already been modified by an adjective. If the model represents this crucial part of the structure correctly, but makes mistakes elsewhere in the trees, you should grade the model as being correct.

03 Movement task

The files that contain the replies of the models you will evaluate are located in the shared `03_movement` folder.

In this experiment, we ask the model to complete the following task:

Consider the sentence below. Provide code for a syntactic tree representing its structure that can be rendered with LaTeX's forest package. Assume X-bar theory. Pay special attention to syntactic movement and represent all instances of movement with (indexed) traces. Do not write code to draw the actual syntactic arrows that connect the traces.

Your job is to evaluate the models' performance on these tasks and to mark the code they have produced as either correct or incorrect in the spreadsheet I shared with you. If the performance of a model is satisfactory, you should enter "1" in the corresponding cell for the task. If the performance is unsatisfactory, you should enter "0."

There are four language models you will be evaluating. Correspondingly, in the folder `03_movement`, there are four text files that contain the replies of the four models, named **respectively**: `gpt-3.5-turbo-012.txt`, `gpt-4-0314.txt`, `meta.llama3-1-405b-instruct-v1.txt`, and `o1-preview-2024-09-12.txt`. Each model completed 30 tasks. As such, in total, you will be evaluating $4 \times 30 = 120$ replies. Each of you will be evaluating the same exact outputs of the four models. Nonetheless, in order to ensure your evaluations are independent of each other, you need to work separately.

The spreadsheet I shared with you has four tabs. Each tab corresponds to one of the four models you will be evaluating: gpt-3.5-turbo-012, gpt-4-0314, meta.llama3-1-405b-instruct-v1, and o1-preview-2024-09-12.

The models should include LaTeX code in their answers, which should use the syntax of the `forest` package. You can compile the code provided by the model in a new LaTeX project you create for yourself, copy-paste the forest environment to [this shared LaTeX document](#), or make a copy of the shared LaTeX document for yourself and paste it there.

You should focus only on whether the model correctly represents the one type of movement we aim to test for each sentence. Nonetheless, if the tree is completely jumbled up, categorically wrong, or just overall egregious, you should grade the answer as “0.” The model will often provide the tree code with the opening and closing statements for the forest environment, i.e.:

```
\begin{forest}
  \\ THE CODE FOR THE TREE HERE
\end{forest}
```

If the model doesn’t do that, please add statements `\begin{forest}` and `\end{forest}` where necessary. If the model’s LaTeX code doesn’t compile at all (even though you did everything right, included all the necessary statements, packages, etc.), you should also grade the reply as “0.”

I have shared with you a Google doc *03_movement dataset +trees* that contains the replies of all the models, i.e. it contains the same data as the `gpt-3.5-turbo-012.txt`, `gpt-4-0314.txt`, `meta.llama3-1-405b-instruct-v1.txt`, and `o1-preview-2024-09-12.txt`. If you're the first person to work on a part of a task, please copy-paste the LaTeX trees you will have compiled and paste them right after the corresponding LaTeX code outputted by the model. In this way, maybe we can avoid every single person having to compile the trees separately. While grading has to be done individually, the compiling of LaTeX trees can be a collaborative effort.

Note that the items in the text files as well as the Google Doc are out of order. For example, for the GPT models, item 16 is followed by items 16, 18, 14, etc... (Meta Llama’s answer comes in a yet different order.) I reordered the rows in the spreadsheets correspondingly. Make sure to enter grades into the corresponding correct cells in your spreadsheet.

All the test sentences that the models will be evaluated on are given below. A headline that starts with “#” indicates the intended type of movement.

```
# A-MOVEMENT: PASSIVE VOICE
"The fence had been painted.",
"A tree is being observed.",
```

```
# POLAR AUX-QUESTION
```

"Have you thought of using a second monitor?",
"Can I actually survive this?",

CONDITIONAL AUX-INVERSION

"Were I good at coding, this would be much easier.",
"They would have won had they trained harder.",

NEGATIVE/ONLY AUX-INVERSION

"Never have I been so excited!",
"Only now do I recognize him.",

WH-QUESTION

"Who did they invite here?",
"Why did they let this weirdo enroll?",

WH-QUESTION WITH PIED-PIPING

"What color is the roof of his house?",
"In what way is he a better candidate?",

WH-QUESTION WITH P-STRANDING

"Who did she vote for in the last election?",
"What do they believe in?",

WH-QUESTION PAST C-BOUNDARY

"When do you think she will arrive?",
"What did he say he was going to prepare?",

TOPICALIZATION $\bar{A}$ -FRONTING

"The red book, I have studied assiduously.",
"For his teachings, I have a lot of respect.",

LOCATIVE FRONTING

"Right behind me stood the Archangel Gabriel.",
"Under the bed lay a little black kitten.",

RAISING ADJECTIVE

"Natasha was likely to win the election.",
"The hurricane is certain to wreak havoc.",

RAISING TO SUBJECT

"Mark seemed to be completely losing it.",
"Luna tends to disagree with me.",

TOUGH-MOVEMENT

"Not all life circumstances are easy to alter.",
"Radu Jude's latest film was a delight to watch.",


```
# RAISING TO OBJECT
"Susan believes her niece to be the right pick.",
"I expected the train to be on time.",

# MOVEMENT IN RELATIVE CLAUSES
"The child who I gave a lollipop is smiling.",
"She knows the rough area where he went."
```

Thus, for example, when evaluating the model's performance on the sentence "Natasha was likely to win the election, " you will grade the model on whether it correctly represents the movement of the subject past the raising adjective. If the model represents this particular instance of movement correctly, but makes mistakes elsewhere in the tree, you should grade the model as being correct.

The prompt asks the model to not write code to draw the actual syntactic arrows that connect the traces. This part is added so that the model does not attempt to draw the arrows because the arrow code often turns out to be buggy. Nonetheless, if the model adds arrows that explicitly show the movement do not mark it down for doing so.

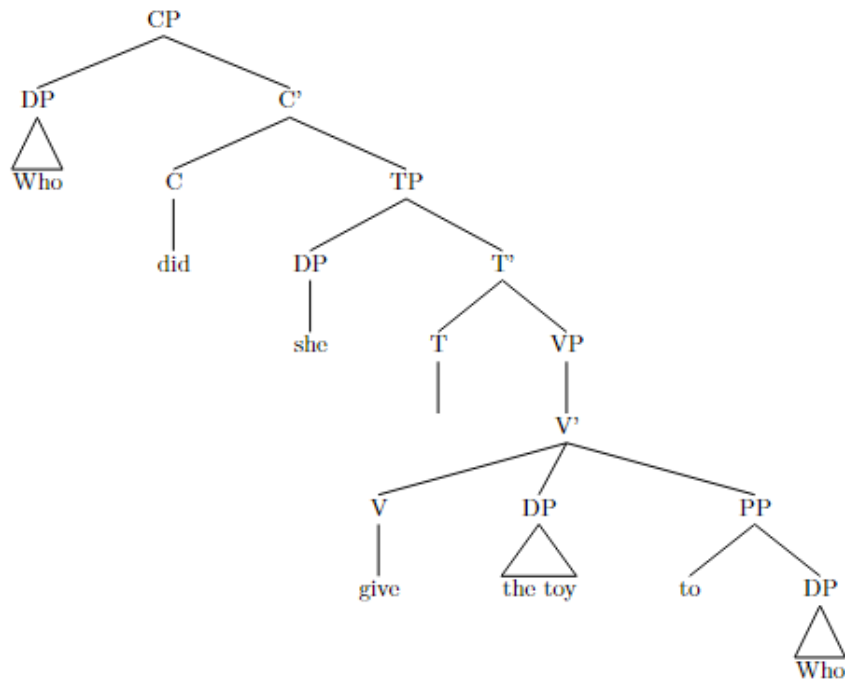
Below I'm listing some examples of what could qualify as the models' good and bad performance to provide a little bit of guidance on how to grade them.

Examples of good answers

Good answer 1

```
PROMPT: Consider the sentence below. Provide code for a syntactic
tree representing its structure that can be rendered with LaTeX's
forest package. Assume X-bar theory. Pay special attention to
syntactic movement and represent all instances of movement with
(indexed) traces. Do not write code to draw the actual syntactic
arrows that connect the traces.
"Who did she give the toy to?"
```

This test sentence is in the category WH-QUESTION. The model outputs code that compiles to:



There are no arrows representing movement, and aspects of the tree are questionable, but two copies of “Who” are present in the right positions. As such, this answer qualifies as correct and should be graded as “1.”

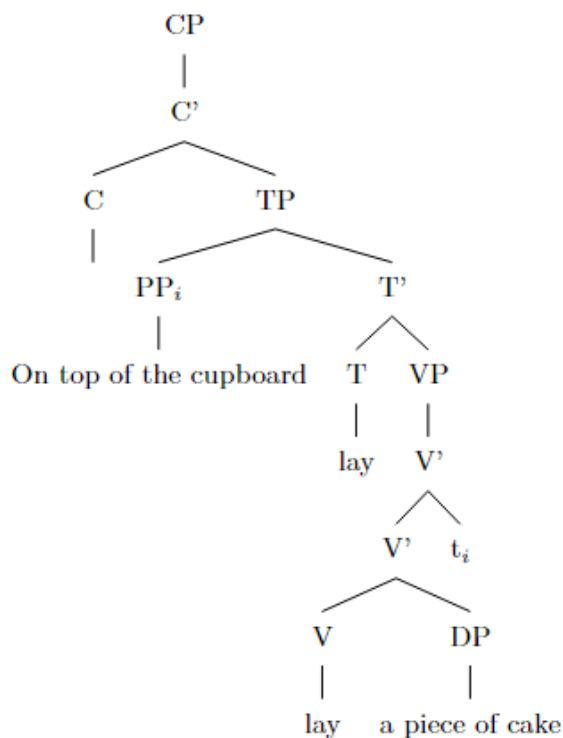
The model represents V' as ternary-branching. Ternary branching is generally assumed to not exist in Minimalist frameworks. Nevertheless, this task belongs in the category `WH-QUESTION`. As such, you’re not grading the model on the structure of VP, and should not mark it down for making this mistake.

Good answer 2

PROMPT: Consider the sentence below. Provide code for a syntactic tree representing its structure that can be rendered with LaTeX's forest package. Assume X-bar theory. Pay special attention to syntactic movement and represent all instances of movement with (indexed) traces. Do not write code to draw the actual syntactic arrows that connect the traces.

"On top of the cupboard lay a piece of cake."

This test sentence is in the category `LOCATIVE FRONTING`. The model outputs code that compiles to:



In the above tree, the locative expression is represented as originating in VP and fronting to the subject position in TP. This is correct. As such, the answer should be graded as “1.”

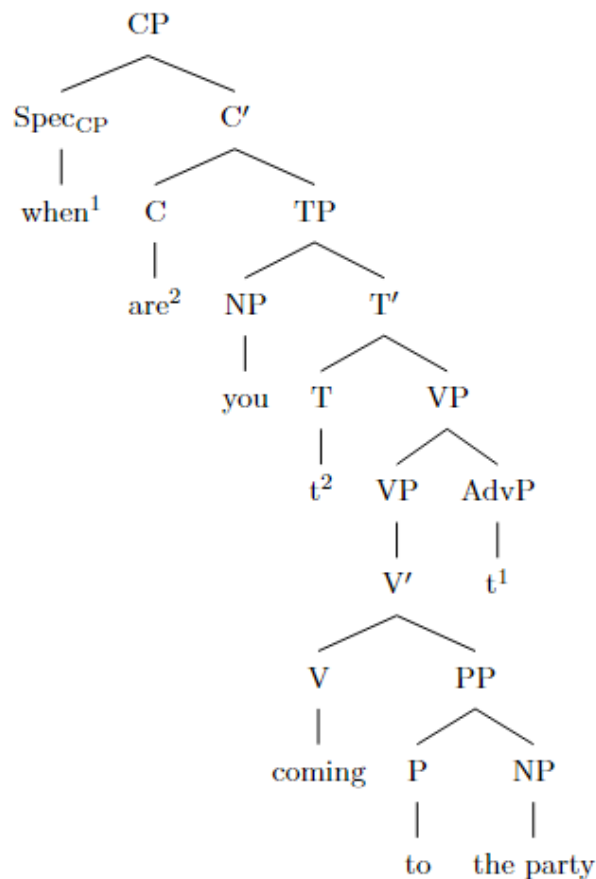
The model also shows that the verb moves from V to T (or is copied and realized twice; the model does not represent the lower copy as a silent trace). V-to-T movement is generally not assumed for English, so this part of the tree could be considered incorrect. Nevertheless, in this task, we’re interested only in `LOCATIVE FRONTING`. As such, you should not mark the model down for making this mistake.

Good answer 3

PROMPT: Consider the sentence below. Provide code for a syntactic tree representing its structure that can be rendered with LaTeX's forest package. Assume X-bar theory. Pay special attention to syntactic movement and represent all instances of movement with (indexed) traces. Do not write code to draw the actual syntactic arrows that connect the traces.

"When are you coming to the party?"

This test sentence is in the category `WH-QUESTION`. The model outputs code that compiles to:



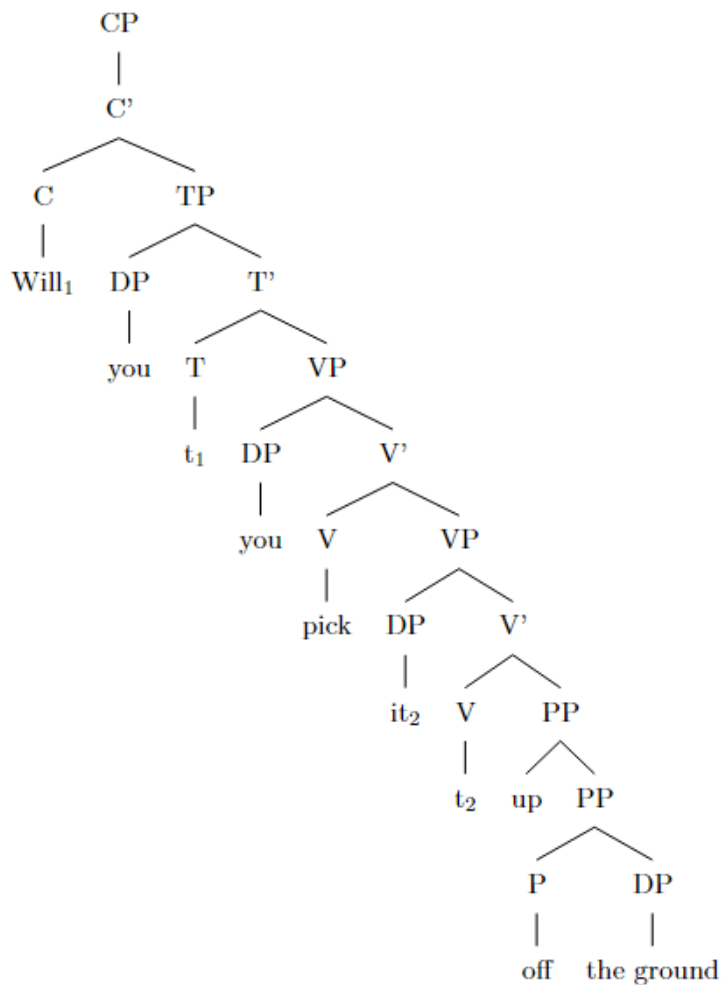
The model correctly represents the movement of the adverbial wh-word from VP to Spec,CP. As such, the answer should be graded as “1.”

Good answer 4

PROMPT: Consider the sentence below. Provide code for a syntactic tree representing its structure that can be rendered with LaTeX's forest package. Assume X-bar theory. Pay special attention to syntactic movement and represent all instances of movement with (indexed) traces. Do not write code to draw the actual syntactic arrows that connect the traces.

"Will you pick it up off the ground?"

This test sentence is in the category POLAR AUX-QUESTION. The model outputs code that compiles to:



The movement of the auxiliary verb to from T to C is correctly represented. There is a lot of funky stuff happening inside the VP, but that is not what we're testing the model here for. As such, this answer can be considered correct; this is graded as "1."

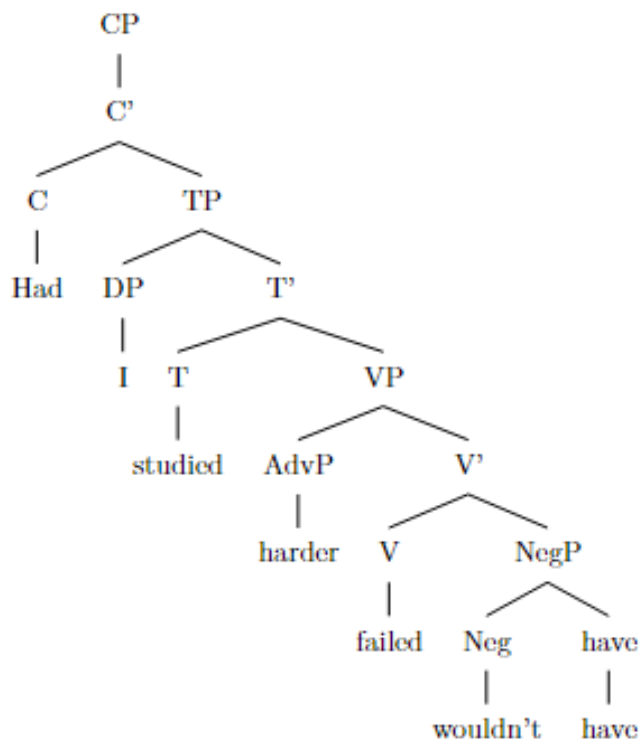
Examples of bad answers

Bad answer 1

PROMPT: Consider the sentence below. Provide code for a syntactic tree representing its structure that can be rendered with LaTeX's forest package. Assume X-bar theory. Pay special attention to syntactic movement and represent all instances of movement with (indexed) traces. Do not write code to draw the actual syntactic arrows that connect the traces.

"Had I studied harder, I wouldn't have failed."

This test sentence is in the category `CONDITIONAL AUX-INVERSION`. The model outputs code that compiles to:



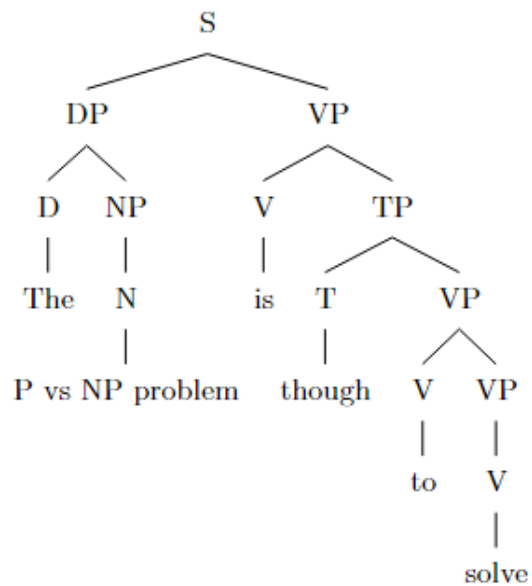
In the above tree, the T-to-C movement of “had” is not indicated at all. As such, the solution provided by the model is incorrect and should be graded as “0”. Also the tree isn’t even complete.

Bad answer 2

PROMPT: Consider the sentence below. Provide code for a syntactic tree representing its structure that can be rendered with LaTeX's forest package. Assume X-bar theory. Pay special attention to syntactic movement and represent all instances of movement with (indexed) traces. Do not write code to draw the actual syntactic arrows that connect the traces.

"The P vs NP problem is though to solve."

This test sentence is in the category `THOUGH-MOVEMENT`. The model outputs code that compiles to:



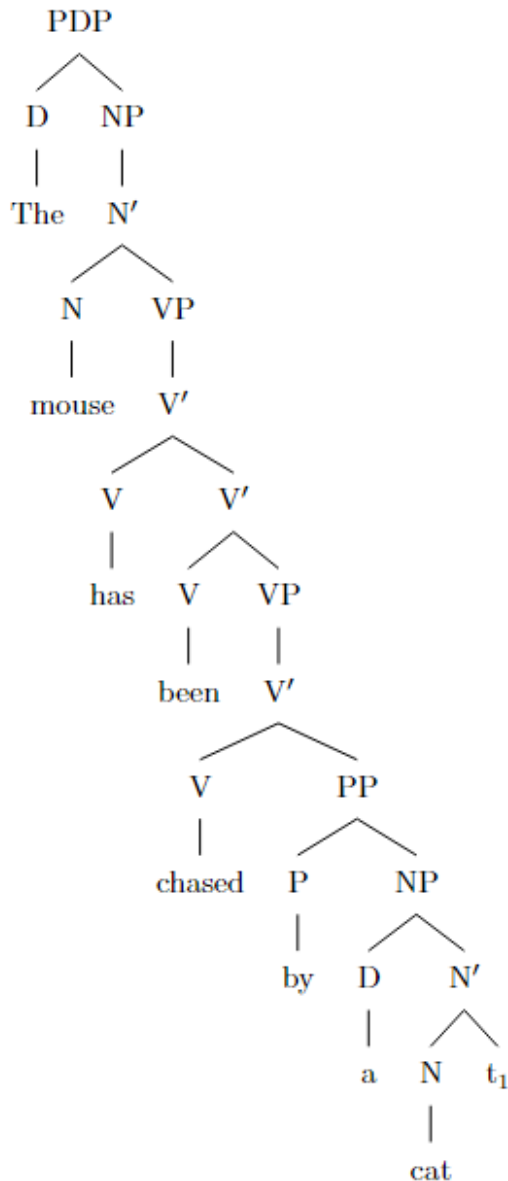
In the model's tree, no movement is represented. As such, the answer should be given a "0."

Bad answer 3

PROMPT: Consider the sentence below. Provide code for a syntactic tree representing its structure that can be rendered with LaTeX's forest package. Assume X-bar theory. Pay special attention to syntactic movement and represent all instances of movement with (indexed) traces. Do not write code to draw the actual syntactic arrows that connect the traces.

"The mouse has been chased by a cat."

This test sentence is in the category A-MOVEMENT: PASSIVE VOICE. The model outputs code that compiles to:



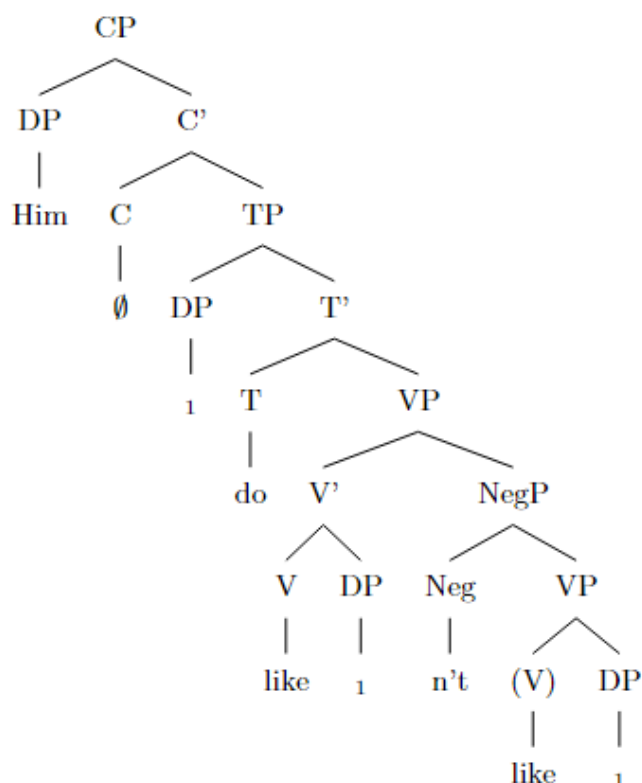
The tree structure is completely jumbled up. There is a trace t_1 , but in the wrong place in the structure. (The trace should stand for the object of the main verb *chased*; it should not be inside the PP.) As such, the answer should be given a “0.”

Bad answer 4

PROMPT: Consider the sentence below. Provide code for a syntactic tree representing its structure that can be rendered with LaTeX's forest package. Assume X-bar theory. Pay special attention to syntactic movement and represent all instances of movement with (indexed) traces. Do not write code to draw the actual syntactic arrows that connect the traces.

"Him, I don't like."

This test sentence is in the category TOPICALIZATION $\bar{A}$ -FRONTING. The model outputs code that compiles to:



In this tree, trace 1 represents the movement of the topicalized “him” through the subject’s Spec,TP position. This is incorrect. As such, this answer gets graded as “0.”

04 Phonology task

Conlang data set construction

When creating phonological datasets, pay attention to whether the dataset is supposed to be neutralizing:

3	neutralizing?	no	no	no	yes	yes	yes	yes
---	---------------	----	----	----	-----	-----	-----	-----

Non-neutralizing rules preserve all contrasts. This is to say, surface forms created by the application of the rule only exist as a consequence of the application of the rule; they do not appear in other contexts. For example, if the rule “pre-glottal (h ɦ ?) nasalization of vowels” is non-neutralizing, then nasal vowels will not appear in the data set except before glottal consonants. If the rule “pre-glottal (h ɦ ?) nasalization of vowels” is neutralizing, then all the vowels appearing before glottal consonants will be nasal, but there will be nasal vowels appearing in other environments as well.

The models will be instructed to focus only on the phones designated as “phones of interest”. As such, make sure that each word you create demonstrates the application or non-application of the rule. For example, if you’re working on the rule “intervocalic spirantization of voiced stops” where the relevant phones are “*b d g > β ð γ*,” each word should be:

1. either something like *saβa, moða, soye, suβum* — demonstrating the application of the rule,
2. or *bap, sab, gapa, pagsen, pamga* — demonstrating the non-application of the rule
3. or *iðid, biβid* — demonstrating both

but we don’t need words like *tap, it, kapa, litu*, which don’t demonstrate either.

In general, always make sure to demonstrate the application and non-application of any rule in all the relevant environments with at least a few example words. Make sure that the dataset you create not only guides the model towards the correct solution by demonstrating the application of the rule in the appropriate contexts, but also successfully rules out all the other possible analyses. Once again, for example, given the non-neutralizing rule “intervocalic spirantization of voiced stops,” make sure that your dataset contains a variety of examples such as:

1. *geblam, mirga* — ruling out the analysis of the dataset as “spirantization of voiced stops between liquids,”
2. *ladbi, pazgak* — ruling out the analysis of the dataset as “spirantization of voiced stops between voiced phonemes,”
3. *panga, raβamban* — ruling out the analysis of the dataset as “spirantization of voiced stops between sonorants phonemes,”
4. *husbar, paʔdug* — ruling out the analysis of the dataset as “spirantization of voiced stops word-internally,” etc.

When checking other people’s datasets, also ask yourself the question: Are all the alternatives ruled out correctly? Is there at least a couple of examples ruling out the alternatives?

Model evaluation

The files that contain the replies of the models you will evaluate are located in the shared 04_phonology folder.

In this experiment, we ask the model to complete the following task:

Below, you are given a list of 40 words, which are the surface forms in a language you have not previously encountered. Each surface form is given as a string of phones separated by spaces. What phonological process operating in this language can you observe? Focus on the following phones of interest: [PHONES OF INTEREST]

State the phonological process as a rule such as e.g. $A \rightarrow B / _ C$, $A \rightarrow B / C _$, or $A \rightarrow B / C _ D$, where A stands for the underlying phoneme, B for its surface realization, and C (and D) specify the environment where A is realized as B. State your rule in the most

general possible terms, i.e. refer to natural classes whenever possible (instead of simply listing the affected phonemes).

Your job is to evaluate the models' performance on this task and to mark the (aspects of the) phonological rule it has come up with as either correct or incorrect in the spreadsheet I shared with you. If the performance of a model is satisfactory, you should enter "1" in the corresponding cell for the task. If the performance is unsatisfactory, you should enter "0."

There are four language models you will be evaluating. Correspondingly, in the folder 04_phonology, there are four text files that contain the replies of the four models, named respectively: gpt-3.5-turbo-012.txt, gpt-4-0314.txt, meta.llama3-1-405b-instruct-v1.txt, and o1-preview-2024-09-12.txt.

Each model completed 30 tasks. As such, in total, you will be evaluating $4 \times 30 = 120$ replies. Each of you will be evaluating the same exact outputs of the four models. Nonetheless, in order to ensure your evaluations are independent of each other, you need to work separately.

The spreadsheet I shared with you has four tabs. Each tab corresponds to one of the four models you will be evaluating: gpt-3.5-turbo-012, gpt-4-0314, meta.llama3-1-405b-instruct-v1, and o1-preview-2024-09-12.

The models are not asked to provide LaTeX code in their answers, but they still do that sometimes. The LaTeX code should be for the most part human legible, so it's possible you won't have to compile it. If you'd like to compile the code, you can do so in a new LaTeX project you create for yourself, copy-paste the LaTeX bit to [this shared LaTeX document](#), or make a copy of the shared LaTeX document for yourself and paste it there.

The intended corrected solutions to the problems are given below. (Note that they are not stated as rules; the models will/should be using the rule format, but do not mark them down for describing the phonological process in words.)

1. word-final devoicing of obstruents
2. nasalization of vowels after nasal consonants
3. pre-glottal (h ɦ ʔ) nasalization of vowels
4. pre-front vowel (i, e, æ) palatalization of velars
5. intervocalic spirantization of voiced stops
6. vowel (a e i o u) lengthening before vowels or approximants (a e i o u l r w j)
7. post-front vowel (i, e, æ) palatalization of alveolars
8. pre-nasal vowel (ã ĕ ĭ õ ũ) nasalization of voiced stops
9. mid vowel laxing before consonants
10. post-vocalic preaspiration of voiceless stops
11. fricativization of oral stops before oral stops
12. postnasal voicing of fricatives
13. word-final raising of low vowels
14. dentalization of alveolars after front vowels (i e ε æ)
15. preglottalization of voiceless stops after vowels
16. uvularization of velars before back (o u ɔ ɑ) vowels

17. post-labial consonant (m b v p f) rounding of vowels
18. labialization of consonants (^w) after rounded vowels (o y u ø ʏ ʊ)
19. devoicing of vowels between voiceless consonants
20. susurrant after voiced obstruents
21. word-final devoicing of obstruents
22. nasalization of vowels after nasal consonants
23. raising of mid vowels before voiced consonants
24. pre-vocalic aspiration of voiceless stops
25. word-final voicing of obstruents
26. denasalization of vowels after nasal consonants
27. high vowel lowering after laterals (l ʎ ɭ ʟ)
28. gemination of fricatives between vowels
29. word-final voicing of obstruents
30. denasalization of vowels after nasal consonants

You will give a separate grade for three aspects of the rule: *input*, *output*, and *environment*. The *input* grade corresponds to whether the model correctly identified the input of the rule. The *output* grade corresponds to whether the model correctly identified the output of the rule. The *environment* grade corresponds to whether the model correctly identified the environment of the rule. For each component, the model's performance can be graded as either correct ("1") or incorrect ("0").

The models perform better when all the characters are separated by spaced (and as such tokenized separately). For this reason, spaces have been inserted between all the characters in the words in the datasets.

Although the instructions say that the model should provide a phonological rule, do not mark a model down if the phonological process is correctly described in words (but the rule is not stated). Similarly, do not mark a model down if the rule is incomplete but the description surrounding it adds the necessary details. Try to assess whether the model "got it" the same way you'd assess whether a student understood what a given phonological problem is (even if they got the formalism wrong). In other words, the primary goal is not to assess the use of formalism but rather whether the model "understood" what's going on.

Below I'm giving some examples of different answers to show what could qualify as a model's good and bad performance to provide a little bit of additional guidance on how to grade them.

Sample problem 1

Phones of interest

s z ʃ ʒ

Dataset

SR	gloss
u z i ʃ	white

a z u 3 uncle
 a r j u 3 proceed
 z u s k a black
 s i z u 3 heavy
 u v i f stream
 a r a 3 try
 i r a f street
 u n s a take
 g a 0 a 3 east
 z a 0 i 3 nobody
 i z u 3 fur
 a s a f some
 z a z i 3 lizard
 v a z u 3 melon
 u g z i f chickpea
 i z a 3 how
 w u d a f coastline
 u t i 3 nail
 z a m i f beaver
 h i 3 cinnamon
 u z a 3 they (dual)
 i z i 3 forefinger
 u z b u 3 duck
 z a n a w strawberry
 a ɲ a p storm
 a s u 3 buy
 ɲ i v u f fresh
 z i n u f discover
 i z i k fog
 i s a f cat
 p u 0 a 3 calf
 k a n z a g bring
 i s i 3 dog
 u 3 grow
 a k u f group
 a m i 3 baby
 n i z i f quit
 ? i k a 3 father
 i 3 love
 u z z i 3 feed
 u s i f apple
 z u s a f those
 z i f believe
 a d a f ant
 t u w a f realize

i b j i ʃ roll
 z a z a ʒ awaken
 a ɔ i ʒ sister
 i z u ʃ eyelash

Target answer

word-final alveolar fricatives

Model's answer

The phonological process observed in this language appears to be one of assimilation in the context of sibilant phonemes (s z ʃ ʒ). The observed behavior indicates that a sibilant becomes voiced when it is placed in proximity to another voiced consonant, generally in the onset or coda of the syllable. Here is the rule that describes this process succinctly:

$$\left[\begin{array}{l} +\text{sibilant}, \\ -\text{voice} \end{array} \right] \rightarrow \left[\begin{array}{l} +\text{voice} \\ \text{---} \end{array} \right] \text{C} +\text{voice}$$

This rule reads as "a voiceless sibilant becomes a voiced sibilant when it is followed by a voiced consonant."

Here are some examples from the provided data to illustrate this rule:

1. /i s i ʒ/ (dog) - The /s/ becomes /ʒ/ due to following /iʒ/.
2. /i s a ʃ/ (cat) - The /s/ stays as /s/ because it is not followed by a voiced consonant.
3. /z i n u ʃ/ (discover) - The /i s i ʒ/ (- /z/ due to preceding /z/).
4. /u z z i ʒ/ (feed) - The rule applies again with /u s z i ʒ/, resulting in /u z z i ʒ/ due to proximity.

This general rule adequately encapsulates the phonological pattern observed among the voiced and voiceless sibilants in the given data set.

Evaluation

The input should be "alveolar fricatives," not "voiceless sibilants"

input: 0

The output should be "postalveolar fricatives"

output: 0

The environment should be “word-finally”

environment: 0

Sample problem 2

Phones of interest

b β d z g ʎ

Dataset

SR	gloss
ʊ z ɛ p	rye
ə z ə p	go
ə ʎ ʊ n	meat
l s	flame
a β ɔ d	onion
ɛ ʎ l g	eye
i β a b	hair
b i β o ʃ	sister
u z l h	red
ʊ z ɛ g	buy
g ə ʎ i g	lentil
ɛ β l g	scorpion
ɛ ʎ l d	prepare
ɔ ʎ ə ŋ	ignore
ɔ ʎ ʊ b	some
tʃ i β o g	kill
b ʊ r b ə	stone
g o dʒ l bhear	
l ʎ ɔ g	repair
g ə ʎ ə	chest
ʊ n n o g	jaw
o β ɛ b	forehead
a r l d	waterfall
ʊ β ɛ d	tell
ɛ ʎ ə p	gravy
ʊ ʎ ɛ w	moon
d i z l ʃ	push
ɔ ʎ u d	lion
t ɔ ʎ u g	shark
ɛ ŋ g a	every
b ɛ z ʊ dʒ snake	
g u β l ŋ	cashew
ə z a d	most

u z ɔ b new
 g ɔ β ɔ b dry
 i ʏ ɪ b family
 r ɔ β ʊ r elephant
 d i z ə ŋ walk
 n ɛ β ə g acquire
 t ə β ɛ d these
 ʊ z ɪ b bird
 t ɪ z a tʃ woman
 g o β ə d know
 i g d ʊ direction
 ɛ ʏ ɛ d fist
 ɛ ʏ ʊ h north
 g ʊ z o day
 b ʊ ʏ ə b blueberry
 a z u n banana
 a z ɔ d wine

Target answer

intervocalic spirantization of voiced stops

Model's answer

The phonological process observed involves the alternation between voiced stops (b, d, g) and their corresponding fricatives (β, ʏ) or the voiced alveolar fricative (z) in specific environments.

The phonological rule can be summarized as:

1. Voiced stops become fricatives between vowels (intervocalic position):

- b → β / V_V
- d → z / V_V
- g → ʏ / V_V

Examples:

- b: a β ɔ d (onion), i β a b (hair), u z ɔ b (new)
- d: g o β ə d (know), t ə β ɛ d (these)
- g: ɛ ʏ ɪ g (eye), g ə ʏ ə (chest), a z ɔ d (wine)

Therefore, the rules can be stated as:

1. b → β / V_V
2. d → z / V_V
3. g → ʏ / V_V

Evaluation

The model states “voiced stops become fricatives between vowels (intervocalic position),” which is correct

input: 1
output: 1
environment: 1

Sample problem 3

Phones of interest

b d g Ø

Dataset

SR gloss
ə a g she
ə u ŋ bathe
d ə assume
u ə g grape
i a b wolf
a i b choose
u ɔ ŋ old
a i ʒ two
u i b none
ʒ i ɔ cold
m ɔ i p seed
u i forehead
ə u n father
d a dʒ i b forest
a ɔ b begin
ʃ i i g kill
d u a quinoa
a ɔ dʒ cloud
ə u finger
ɔ ə ŋ ice
ʃ a u k smoke
ŋ ɔ ə g sea
g ə i g receive
u ə g wife
ɔ z i b boil
p i ə d chicken
ə ɔ yellow
j u dʒ dʒ i d drink
i i b rhinoceros
ɔ u n ear

g ɔ u you (dual)
 ɔ a g make
 ʒ ɐ ɔ w hill
 u ɔ d cod
 b a ɐ g we
 ʒ ɐ a t regret
 u ɐ ʒ there
 w u m i b few
 i u g broken
 w ɐ i n flat
 ɐ ɔ r leave
 ɔ z ɐ f bile
 i ɐ see
 f a ɔ z green
 g ɐ ɔ ʃ dry

Target answer

intervocalic deletion of voiced oral stops

Model's answer

Looking at the provided data set and focusing on the phones of interest (b, d, g, Ø), it seems that these consonants (b, d, g) appear at the beginning of words, between vowels, or at the end of words, but their occurrence suggests a possible pattern related to vowel adjacency. Upon closer examination, there appears to be a tendency where the underlying phoneme 'd' surfaces as 'Ø' (i.e., it is deleted) in certain environments:

Occurrence of /d/:

- d ɐ - assume
- d a d̥ i b - forest
- d u a - quinoa
- p i ɐ d - chicken
- j u d̥ d̥ i d - drink
- u i b - none
- ɔ a g - make

Absence where /d/ might have been expected:

- ɐ u n - father
- ɔ á n - ear
- ɐ ɔ r - leave
- u ɐ - wife (possibly *udua*)
- ɐ u ŋ - bathe (possibly *duŋ*)

From the evidence, the rule appears to be that /d/ is deleted (Ø) when it occurs between two vowels. This leads to the following phonological rule:

/d/ → Ø / V_V

This means that the phoneme 'd' is realized as Ø (i.e., it is deleted) when it occurs between two vowels.

Evaluation

The model correctly identifies that the rule is intervocalic deletion, but restricts it only to “d” (as opposed to correctly identifying all voiced stops as undergoing deletion). As such, the input is incorrect, but the output and the environment are good.

input: 0
output: 1
environment: 1

Sample problem 4

Phones of interest

p t k p^h t^h k^h ?

Dataset

SR	gloss
ð æ p a t ^h	beard
k ə tʃ ə t ^h	dolphin
æ p ^h f ə	wife
t a dʒ v a d	thigh
ð æ dʒ ə p ^h	wolf
? i n æ t ^h	inland
? u p ^h x æ	meat
æ k ^h	discover
ð ə tʃ ə t ^h	aunt
ʎ æ t ^h θ i b	knuckle
i t a k ^h	allow
z u m ʎ u p ^h	blackberry
u p ^h x ə z	vegetable
a p ^h ? a	seed
? i t ^h p u x	warm
ð a t ^h k	evening
p ə m u p ^h	person
i t ^h t a p ^h	new
i n æ p ^h	mustard
i k ^h ? i ?	neck
v ə k ^h k i ?	star

? u t æ s sand
 ʃ i p^h k ʌbdomen
 p ʌ ? i z receive
 u ? t ʌ blue
 ə ? i k^h wet
 i k^h ? ʌ they (dual)
 ʌ d æ p^h jump
 u k i p^h bright
 ə k ə f valley
 ʌ w u p^h regret
 i w u k^h earlobe
 g æ ? u k^h he/she/it
 m æ dʒ i k^h yell
 ə ? ə k^h full
 æ p^h t ə feel
 m i k^h parent
 ə ʏ d ə t^h that
 ? æ ʃ ə k^h monkey
 z u ? ʌ k^h brown
 ʌ t ə t^h adapt
 k u ʒ b æ ? hope
 ʌ p^h k i owl
 ə k^h ? u k^h all
 ʌ t^h ? i t^h lemon
 i p^h ? u ?evil
 u k u p^h realize
 dʒ u k^h ? ə give
 i k æ t^h stretch
 dʒ ə k^h punch
 ? ʌ p i k^h direction
 t ə ? æ tʃ threaten
 θ i t^h k u f root
 w i k^h ʃ ʌwhite
 g ə m ə k^h newt
 u θ u t^h melon
 u ? u t^h much
 i p ʌ t^h bean
 ʌ k ʌ p^h eyelid
 æ t^h ? ə p^h rain
 p u t^h s ə assume
 ʌ d ʌ ? we (dual)
 ? ə k^h t ə t^h child
 p ə k^h p əyou (dual)
 k æ b u t^h corn
 ə k^h ? u t^h one

? a p i p^h lion
 θ æ k^h t i proceed
 ə k^h t æ who
 v u n a p^h group
 p i p^h f a forehead
 a k^h ? a brother
 æ t^h k a b return
 æ p^h t æ v insect
 ? æ p^h t^h ə stream
 æ k i g not
 p u p^h k æ d fail
 p a t^h f ə d flat
 a p^h tongue
 u j æ t^h gravy

Target answer

aspiration of voiceless stops in codas

Model's answer

The observed phonological process in the given language appears to be as follows:

- ****Underlying phoneme:**** /p/, /t/, /k/
- ****Surface realization:**** /p^h/, /t^h/, /k^h/
- ****Condition:**** In syllable-initial position

Thus, the phonological rule can be stated as:

Voiceless stops are aspirated in syllable-initial position:

- /p/ → [p^h] / # _
- /t/ → [t^h] / # _
- /k/ → [k^h] / # _

This indicates that the voiceless stops /p/, /t/, and /k/ become their aspirated counterparts /p^h/, /t^h/, and /k^h/ when they appear at the beginning of syllables.

Evaluation

The model correctly identified the rule of aspiration but in the wrong environment.

input: 1

output: 1

environment: 0

Sample problem 5

Phones of interest

e ε o ɔ

Dataset

SR	gloss
d ³ ε b g	adecide
f u p p ε ŋ	bile
s ɔ g n e	all
ɔ ? ʃ i	teach
θ ε x ʃ e	grow
s ε t f ʊ	ʃwet
? ɔ r ŋ o	knee
d ³ o s ɔ g	neck
b ɔ g b e	leopard
ɔ ŋ d u f	kill
d e t ε n	many
ɔ ŋ t ³ ʊ	few
g ε x p e	these
? ɔ n x u g	forest
a d ε n	understand
e g ɔ n	goose
b ε m x i	none
d ³ ɔ x ɔ	xeel
o x ɔ r	lie down
f ɔ j	feel
ɔ g d ³ e	abdomen
b ɔ x ? ʊ t ³	ant
t o t ³ ε θ	dirt
ε p ɔ k	eyebrow
n ε d ³ d	ohug
e k ɔ ŋ	coyote
ε s t ³ ε s	she
? ɔ ŋ b i	they (neut)
u k ɔ r	street
j e t ɔ x	flame
ɔ d m o	survive
e s ɔ t ³	husband
g ɔ f p ʊ	digest
ε g ŋ e	where
j ε f x ε ?	leave
o ? ɔ p	pink
ŋ ɔ n ɔ x	cut
ε d j ɔ ?	yell

j ʊ n ɔ f rain
x ɛ d̥ b ɔ s i s t e r
f ɔ m ʃ o white
p ɔ s ɛ x m a n
ɛ n f ɛ d̥ d a u g h t e r
ɛ n d̥ i b t o n g u e
j u ʔ ɔ ŋ b l u e
g e x ɛ t̥ c a n t a l o u p e
o θ ɛ p e v e n i n g
o s ɔ x f i s t
t e f ɛ ŋ s k i n
ɛ s s e a
e s ɛ b w e
o r i j v i n e g a r
f o t ɔ f w e s t
f ɛ r t̥ e b a r l e y
b o x u k r o o t
ɛ n ŋ o a g r e e
j ɛ m ɔ k c l o u d
ɛ d̥ j e e l e p h a n t
ɛ n t̥ e p e a n u t
ɛ t̥ s o y b e a n
e k ɔ d̥ b e r r y
ɔ t f i k s p i c y
k ɔ ŋ θ a θ b e a v e r
o f ɔ b y o u (s g)
ɛ ʃ s e o l d
t̥ e f ɔ n a d o r e
θ ɔ ŋ g ɔ d c l e a r
e b ɛ t b r o t h
b ɔ b n a s u g a r
x o d̥ ʊ t t r a d e
ʔ ɛ ʔ s e c a r r o t
ʔ ɔ θ p i ʃ g o o d
e b i θ f u l l
t̥ e ŋ a m n e e d
x e r ɔ d̥ t e n
θ ɛ d̥ n e p a r e n t
ʊ k ɛ ʃ s m o o t h
ɛ x f ʊ s e a r
ŋ ɛ x ʔ o p e a r
ɛ k f e n a m e
ɔ t̥ ʔ o s n a k e
s e f i θ c a r r y
ɔ t x o m e a t

t ɛ ʔ ʔ ʊ face
e dʒ ɔ ʔ resist

Target answer

mid vowel laxing in closed syllables

Model's answer

Based on the provided word list, we can observe a phonological process of vowel alternation. Specifically, the mid-front vowel /e/ realizes as [ɛ] and the mid-back vowel /o/ realizes as [ɔ] between two consonants or in the word-final position. This can be expressed in the following way using the requested rules:

- 1) $e \rightarrow \epsilon / _ C\#$ (where $_$ stands for the position of the phoneme of interest and $C\#$ signifies a consonant in word-final position)
- 2) $o \rightarrow \textcircled{o} / _ C\#$
- 3) $e \rightarrow \epsilon / C_C$ (where C_C signifies a consonant environment on both sides)
- 4) $o \rightarrow \textcircled{o} / C_C$

These rules suggest that the /e/ and /o/ vowels tend to lower to [ɛ] and [ɔ], respectively, when they are adjacent to a consonant or at the end of a word.

Evaluation

The model correctly recognizes the rule of mid-vowel laxing, but incorrectly identifies the environment.

input: 1

output: 1

environment: 0